



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: College Physics II / A Dept.: Physics

Exam Duration: 2 hours Exam Paper Setter: Physics Teaching Team

Question No.	1	2	3	4	5	6	7	8	9	10
Score	33	6	13	13	12	13	10			

This exam paper contains 7 questions and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions (3 points each), there is only one correct answer for each question.

1) The electric field at a distance of 10 cm from an isolated point particle with a charge of 2×10^{-9} C is:

- A) 1.8 N/C B) 18 N/C C) 180 N/C D) 1800 N/C

2) The work required to carry a particle with a charge of $6.0 \mu\text{C}$ from a 5.0 V equipotential surface to a 6.0 V equipotential surface and back again to the 5.0 V surface is:

- A) 0 J B) 1.2×10^{-5} J C) 3.0×10^{-5} J D) 6.0×10^{-5} J

3) A 60 W light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

- A) 3600 C B) 3000 C C) 2400 C D) 1800 C

4) If the amplitude of the electric field in a plane electromagnetic wave is 100 V/m then the amplitude of the magnetic field is:

- A) 3.3×10^{-7} T B) 6.7×10^{-7} T C) 0.27 T D) 8.0×10^7 T

5) A positive point charge moves in the negative x direction, through a uniform magnetic field that is in the negative y direction as shown in Fig 1.1. The magnetic force on the charge is:

- A) in the negative x direction
B) in the positive y direction
C) in the positive z direction
D) in the negative x direction

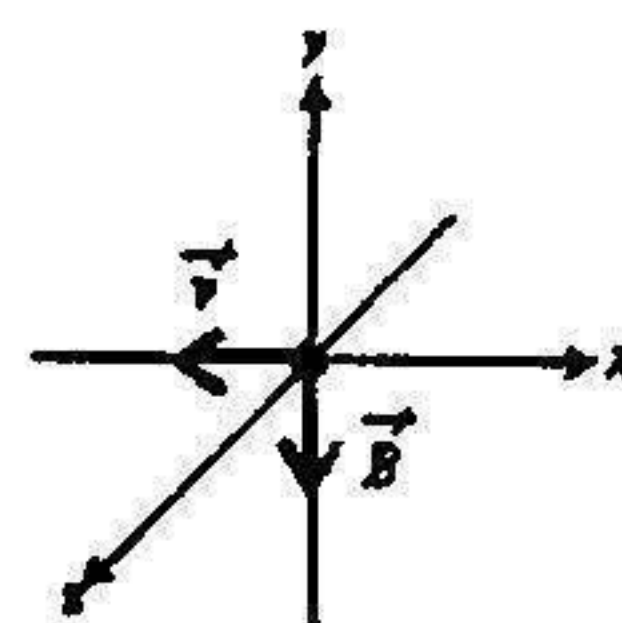


Fig 1.1

6) In Fig 1.2, two long straight wires are perpendicular to the page and separated by distance $d_1 = 0.75$ cm. Wire 1 carries 6.0 A into the page. If the net magnetic field due to the two currents is zero at point P located at distance $d_2 = 1.5$ cm from wire 2, the magnitude of current in wire 2 is:

- A) 2.0 A B) 3.0 A C) 4.0 A D) 5.0 A

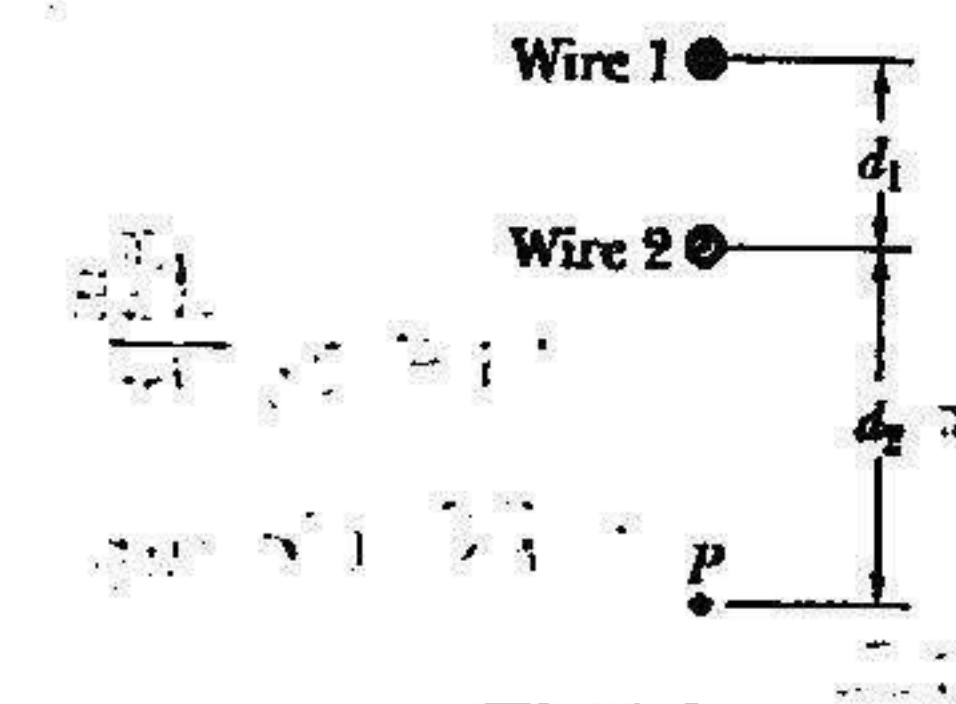


Fig 1.2

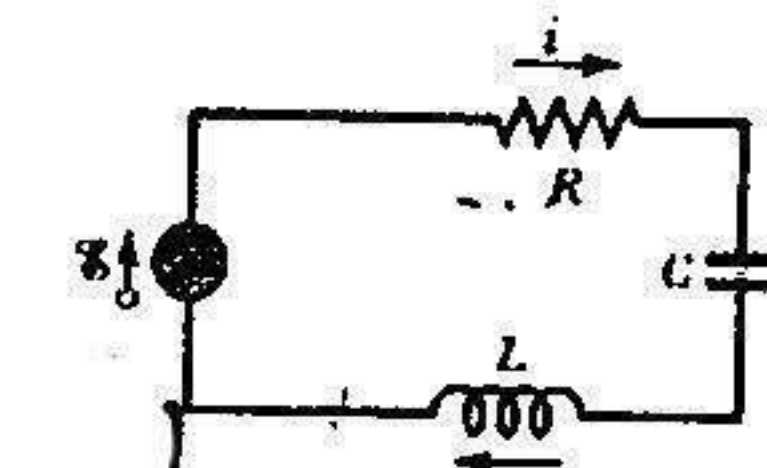


Fig 1.3

7) In Fig 1.3, set $R = 200 \Omega$, $C = 50.0 \mu\text{F}$, $L = 220 \text{ mH}$, $\omega_d = 1000 \text{ rad/s}$ and $\epsilon_m = 36.0 \text{ V}$, the phase angle ϕ is:

- A) -45.0° B) 45.0° C) -60.0° D) 60.0°

8) Fig 1.4 shows a circular region of radius $R = 3.00$ cm in which a uniform electric flux is directed out of the plane of the page. The total electric flux through the region is given by $\Phi_E = (3.00 \text{ mV}\cdot\text{m/s})t$, where t is in seconds. The magnitude of the magnetic field that is induced at radial distances $r = 6.00$ cm is:

- A) $8.85 \times 10^{-20} \text{ T}$
B) $8.85 \times 10^{-18} \text{ T}$
C) 0 T
D) $1.22 \times 10^{-19} \text{ T}$

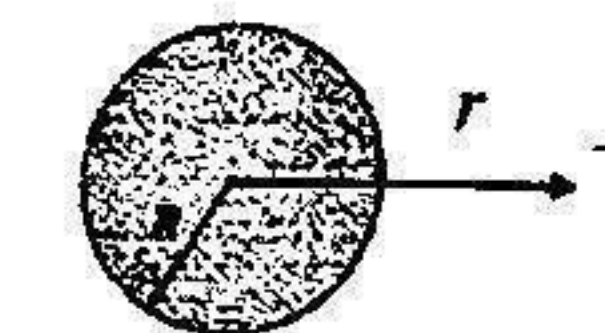


Fig 1.4

9) Light with an intensity $I = 1 \text{ kW/m}^2$ falls normally on a surface and is completely absorbed. The radiation pressure on the surface is:

- A) 1 kPa B) $3 \times 10^{11} \text{ Pa}$ C) $1.7 \times 10^{-6} \text{ Pa}$ D) $3.3 \times 10^{-6} \text{ Pa}$

10) If a certain material's work function ϕ is 2.30 eV and the frequency of the incident radiation is $2.50 \times 10^{15} \text{ Hz}$. The maximum kinetic energy of electrons ejected from the material is:

- A) 6.00 eV B) 8.10 eV C) 10.4 eV D) 12.0 eV

11) Waves with wavelength λ from two slits are in phase at the slits and travel to a distant screen to produce the second minimum of the interference pattern (Just consider interference). The difference in the distance traveled by the wave is:

- A) 0.5λ B) λ C) 1.5λ D) 2λ

Q2. Blank-filling Questions. (Complete each of the following questions according to your calculation)

1. (3 points) The circuit Fig 2.1 shows a parallel-plates capacitor, two ideal batteries, two resistors, and a switch S . Assume $C = 10 \mu\text{F}$, $\mathcal{E}_1 = 10 \text{ V}$, $\mathcal{E}_2 = 30 \text{ V}$, and $R_1 = R_2 = 20 \Omega$. Initially the switch S is open for a long time, then it is closed at $t = 0$, the displacement current through the region between the plates at $t = 5 \text{ ms}$ is A. (Neglect the fringing of the electric field at the edges of the plates)

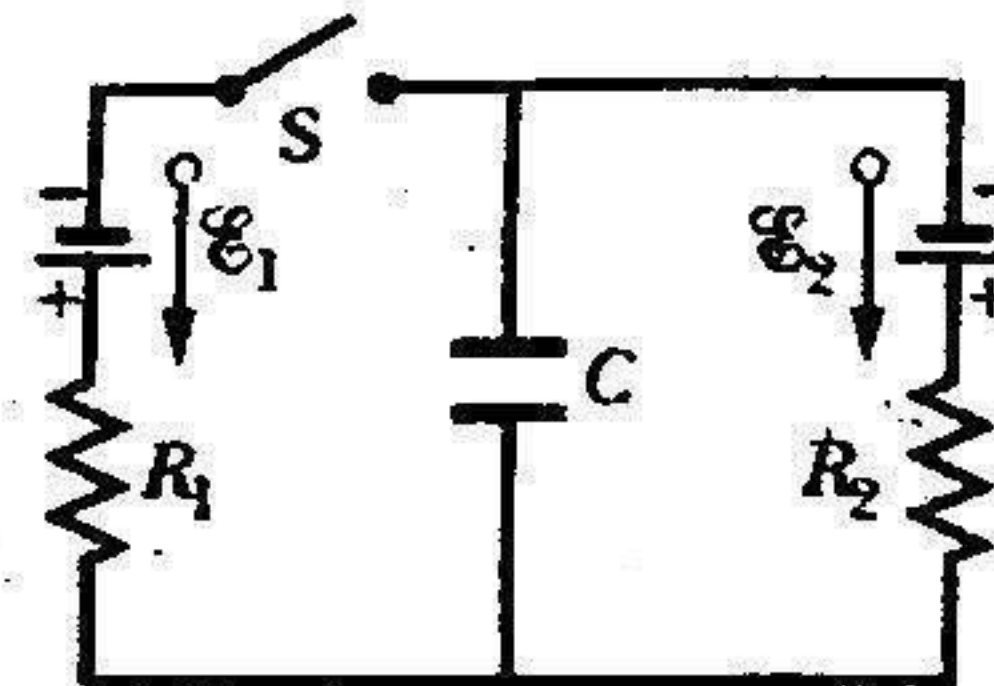


Fig 2.1

2. (3 points) In Fig 2.3, a glass lens is coated on one side with a thin film of magnesium fluoride (MgF_2) to reduce reflection from the lens surface. The index of refraction of MgF_2 is 1.38; that of the glass is 1.50. Assume that the light is incident at angle $\theta = 20^\circ$ to the lens surface, the least coating thickness that eliminates (via interference) the reflections at the middle of the visible spectrum ($\lambda = 550 \text{ nm}$) is nm.

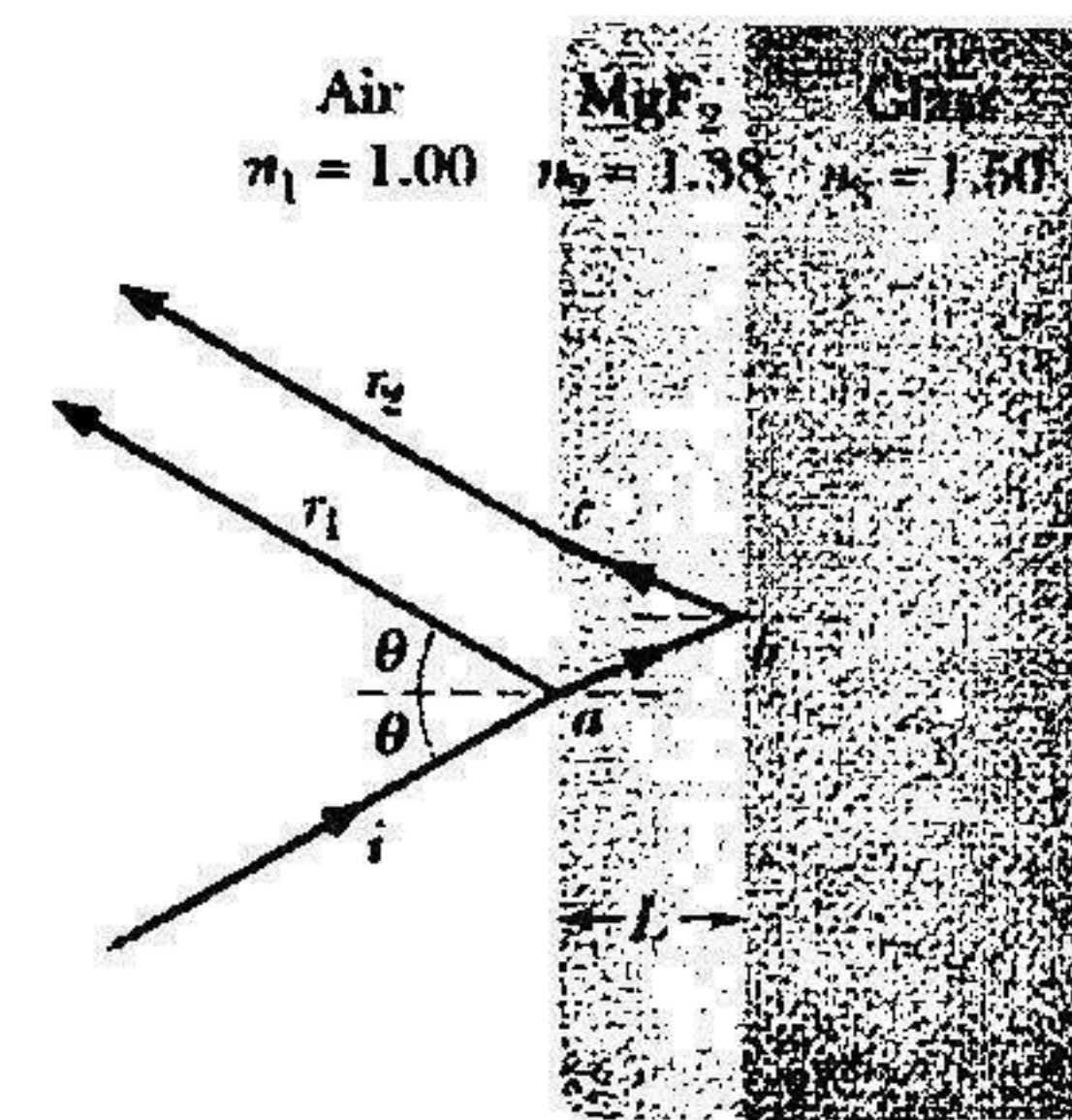


Fig 2.2

Long Questions:

Q3. (13 points) An electron is shot directly toward the center of a large metal plate that has a uniform electric field $E = 1.00 \times 10^5 \text{ V/m}$. If the initial kinetic energy of the electron is $3.80 \times 10^{-17} \text{ J}$ and if the electron is to stop (due to electrostatic repulsion from the plate) just as it reaches the plate.

- What is the magnitude of the electrostatic force on the electron?
- How far from the plate must the launch point be?
- What is the surface charge density of the plate?

Q4. (13 points) In Fig 3, the resistances are $R_1 = 1.0 \Omega$ and $R_2 = 2.0 \Omega$, and the ideal batteries have emfs $\mathcal{E}_1 = 2.0 \text{ V}$ and $\mathcal{E}_2 = \mathcal{E}_3 = 4.0 \text{ V}$.

- What are the size and direction (up or down) of the current in battery 1?
- What is the potential difference $V_a - V_b$?

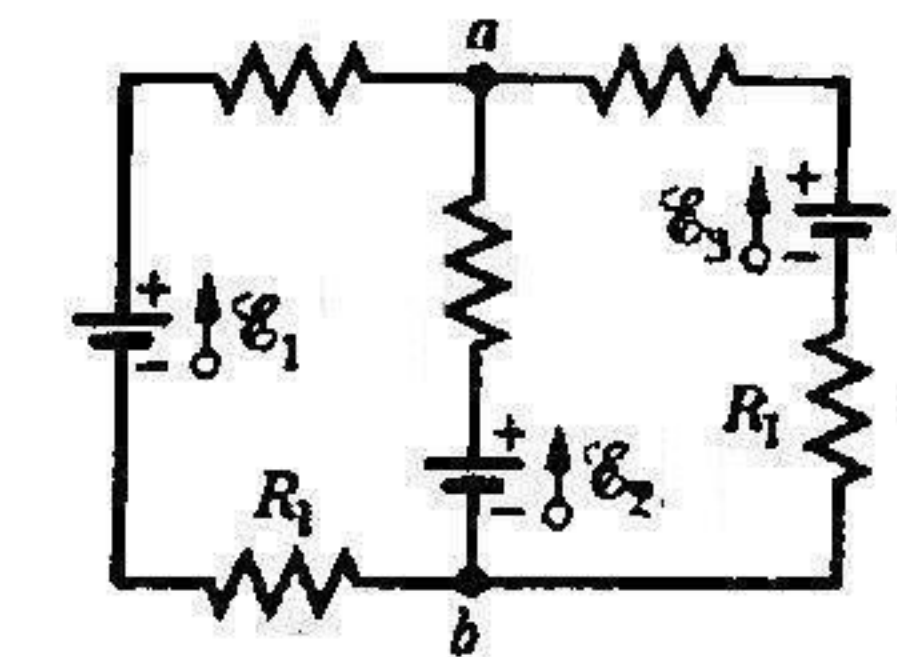


Fig 3

Q5. (12 points) The conducting rod shown in Fig 4 has length L and is being pulled along horizontal, frictionless conducting rails at a constant velocity $\vec{v}$. The rails are connected at one end with a metal strip. A uniform magnetic field, directed out of the page, fills the region in which the rod moves. Assume that $L = 10 \text{ cm}$, $v = 5.0 \text{ m/s}$, and $B = 1.2 \text{ T}$. Assume that the resistance of the rod is 0.40Ω and that the resistance of the rails and the metal strip is negligibly small.

- Find the magnitude and the direction of the current induced in the loop.
- What external force on the rod is needed to maintain constant velocity $\vec{v}$?

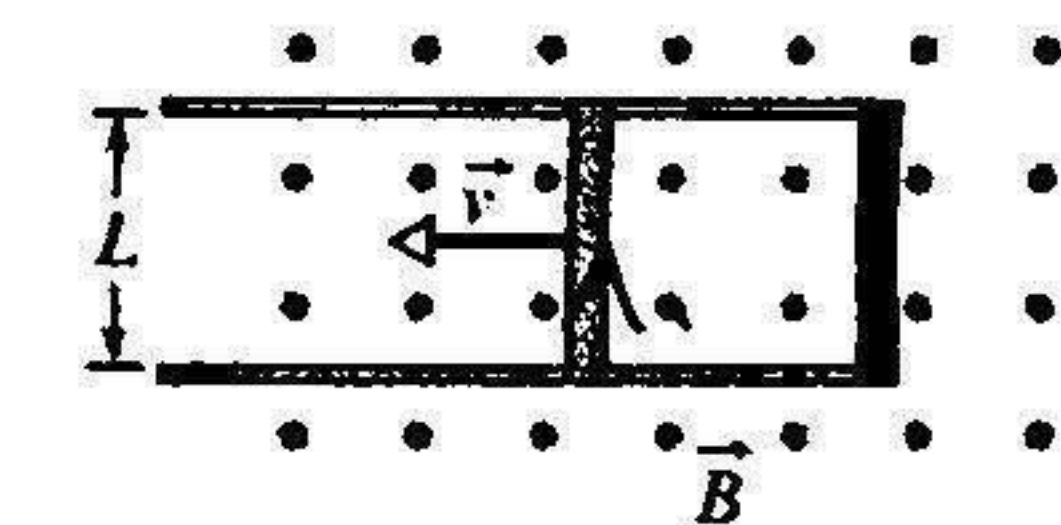


Fig 4

Q6. (13 points) Monochromatic light with wavelength 500 nm is incident on a slit with width $a = 25 \mu\text{m}$. The distance from the slit to a screen is $D = 4.0 \text{ m}$. Let a y axis extend upward along the viewing screen, with its origin at the center of the diffraction pattern. Also let I_P represent the intensity of the diffracted light at point P at $y = 4.0 \text{ cm}$.

- Calculate the ratio of the intensity I_P to the intensity I_m at the central maximum.
- What is the distance on the screen from the center of the diffraction pattern to the first minimum?

7. (10 points) An armada (舰队) of spaceships that is 1.50 ly (light year) long (as measured in its rest frame) moves with speed $0.600c$ relative to a ground station in frame S . A messenger travels from the rear of the armada to the front with a speed of $0.750c$ relative to S . How long does the trip take as measured

- in the rest frame of the armada?
- in the rest frame of the messenger?